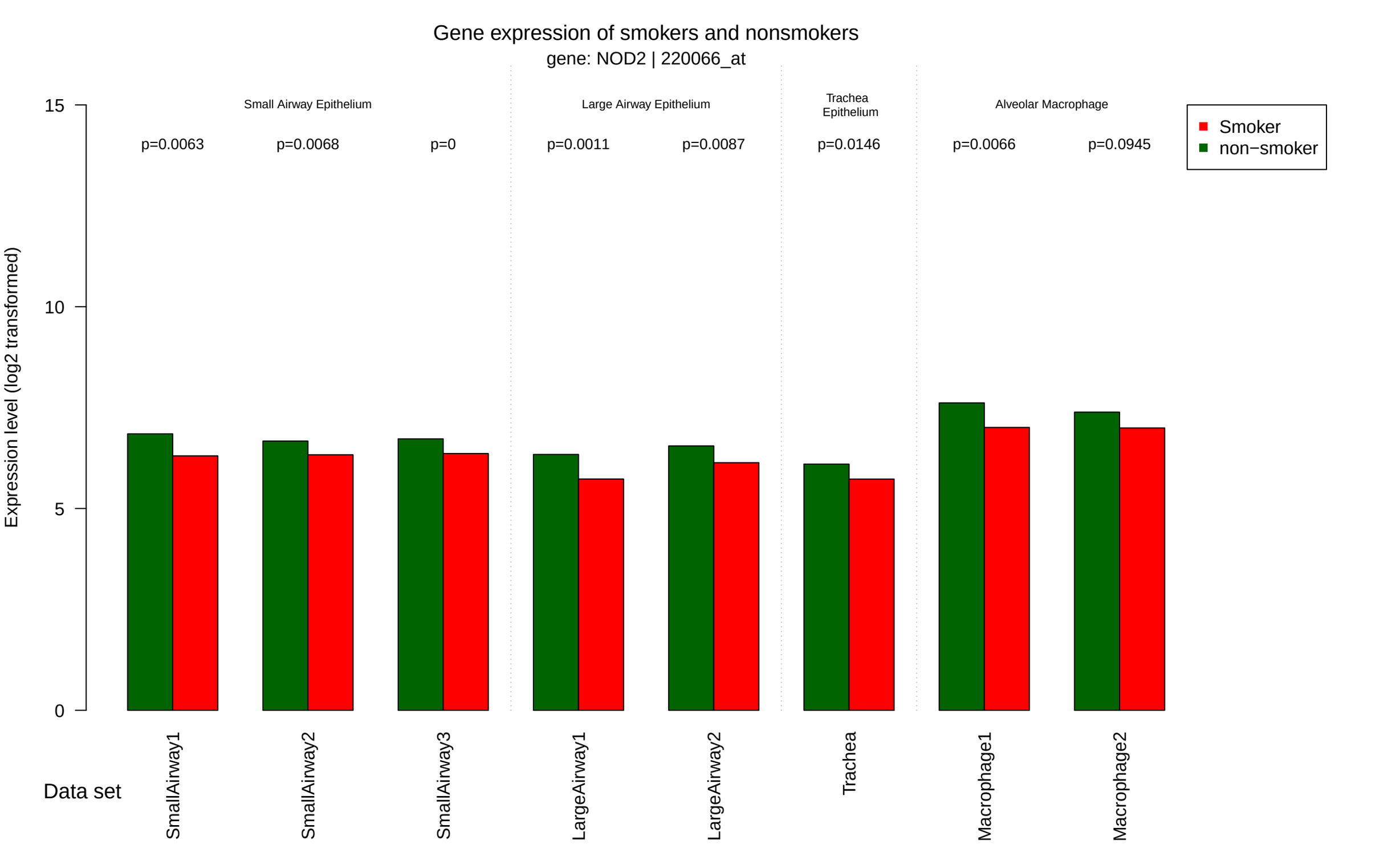
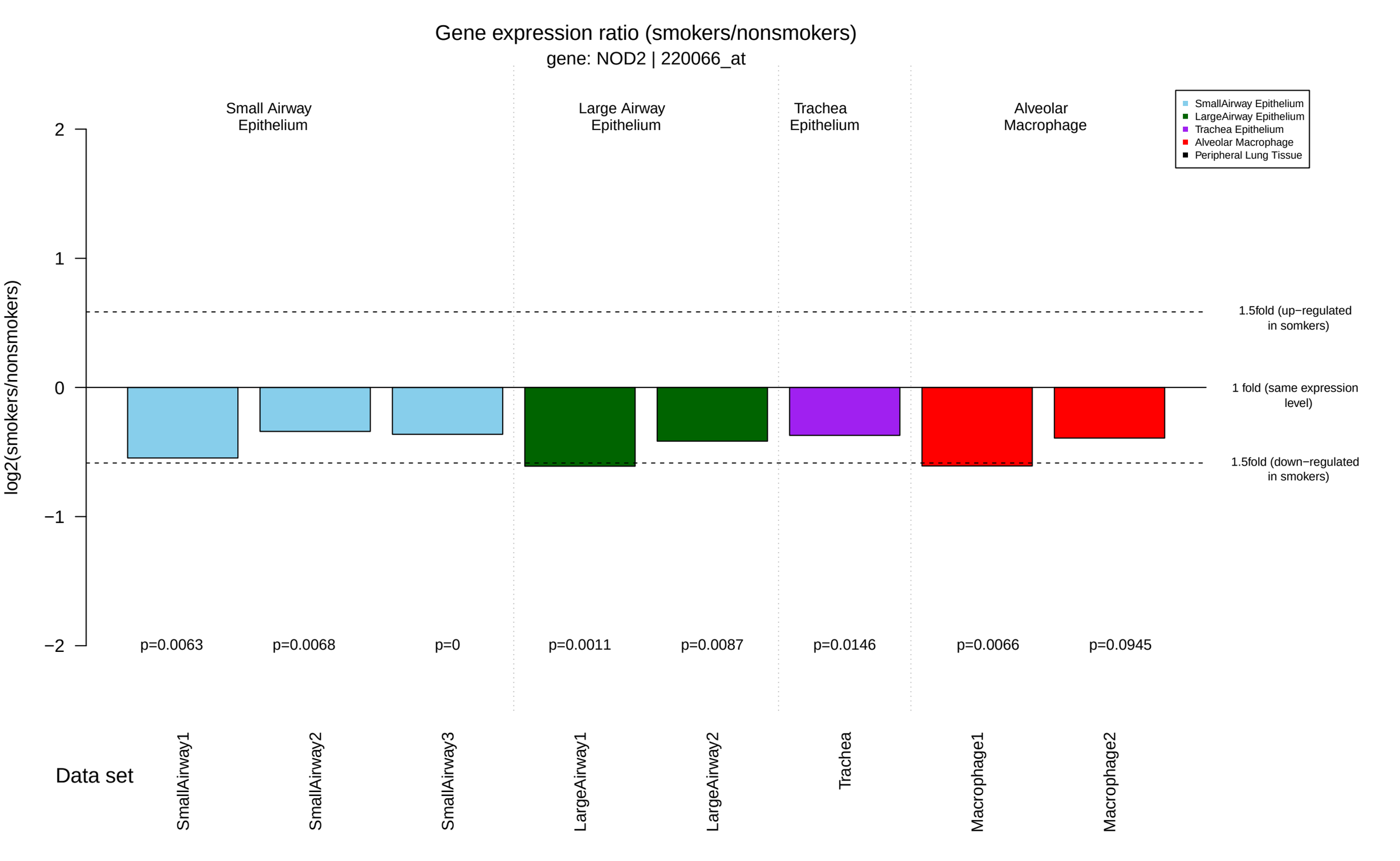
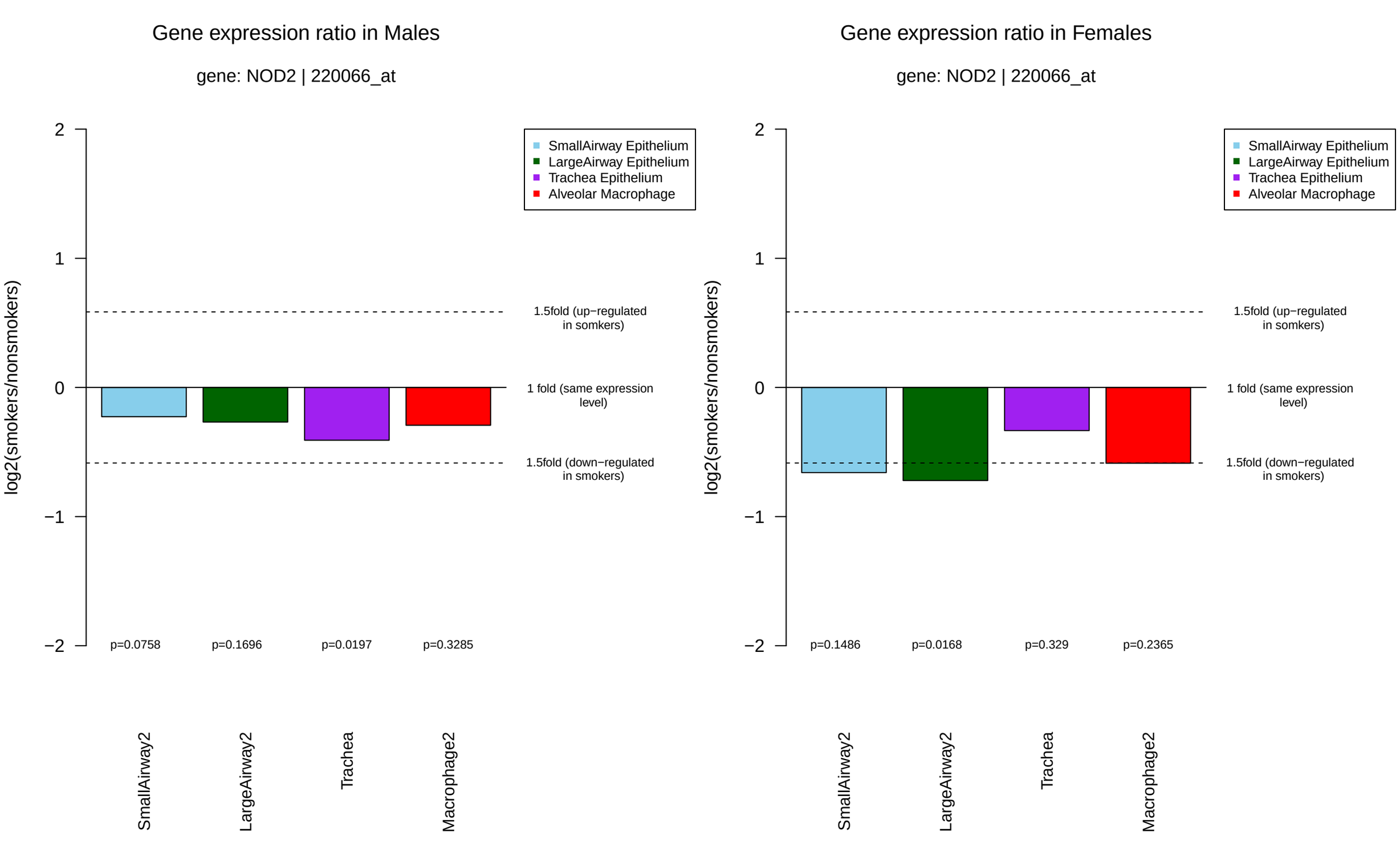
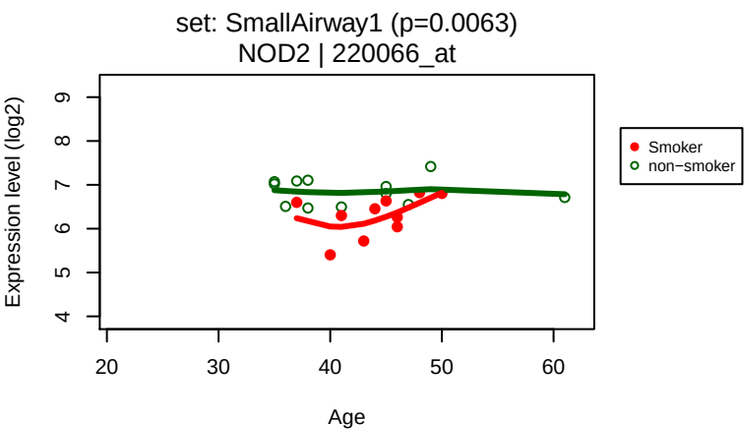
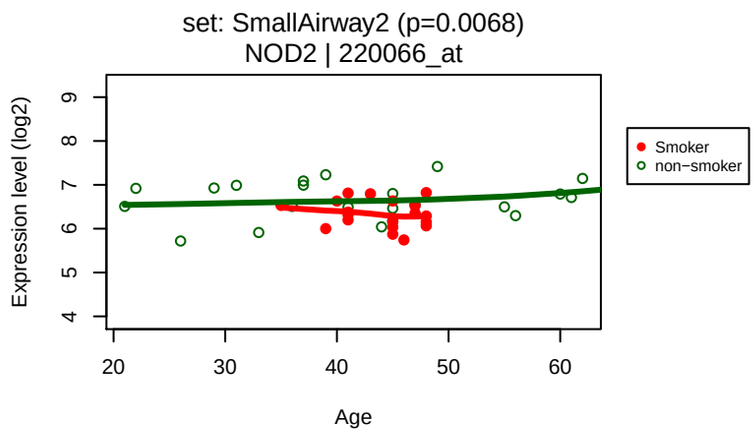
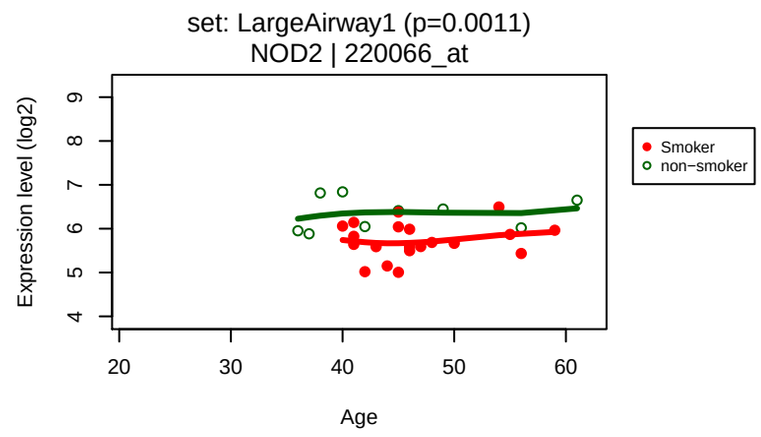

Gene expression of smokers and nonsmokers
gene: NOD2 | 220066_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0063
p=0.0068
p=0
p=0.0011
p=0.0087
p=0.0146
p=0.0066
p=0.0945
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

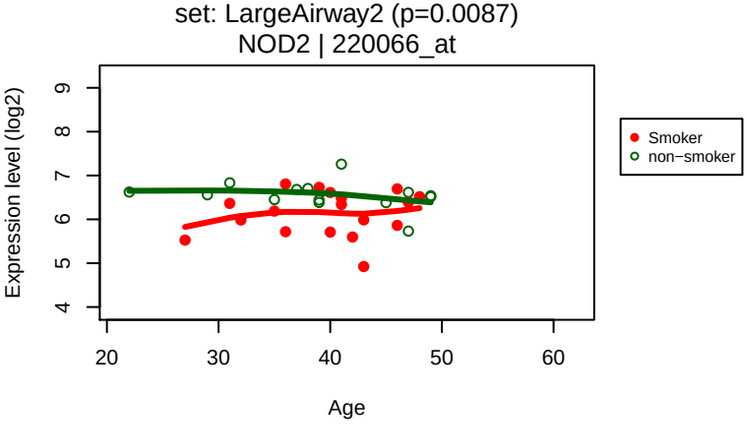

Gene expression ratio (smokers/nonsmokers)
gene: NOD2 | 220066_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0063
p=0.0068
p=0
p=0.0011
p=0.0087
p=0.0146
p=0.0066
p=0.0945
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

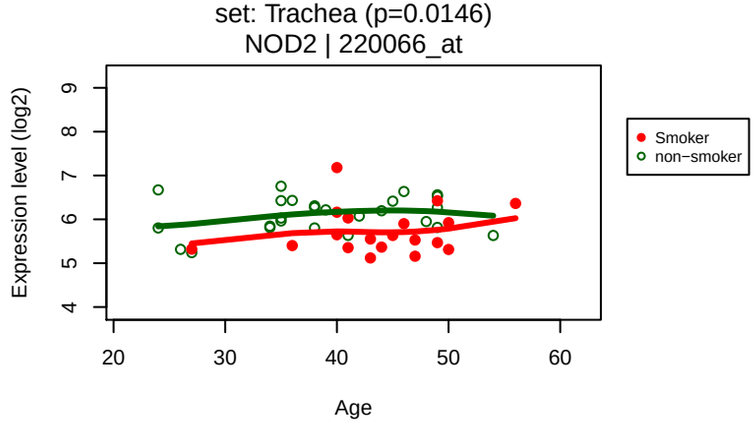

Gene expression ratio in Males
gene: NOD2 | 220066_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0758
p=0.1696
p=0.0197
p=0.3285
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NOD2 | 220066_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1486
p=0.0168
p=0.329
p=0.2365
SmallAirway2
LargeAirway2
Trachea
Macrophage2

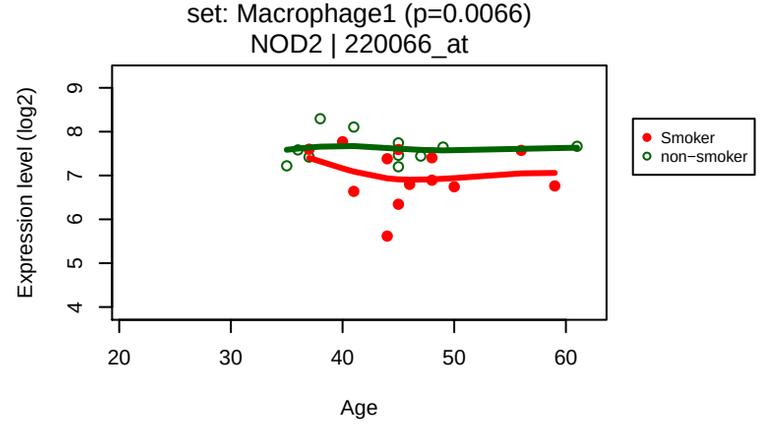

set: SmallAirway1 (p=0.0063)
NOD2 | 220066_at
Expression level (log2)
Age
Smoker
non−smoker

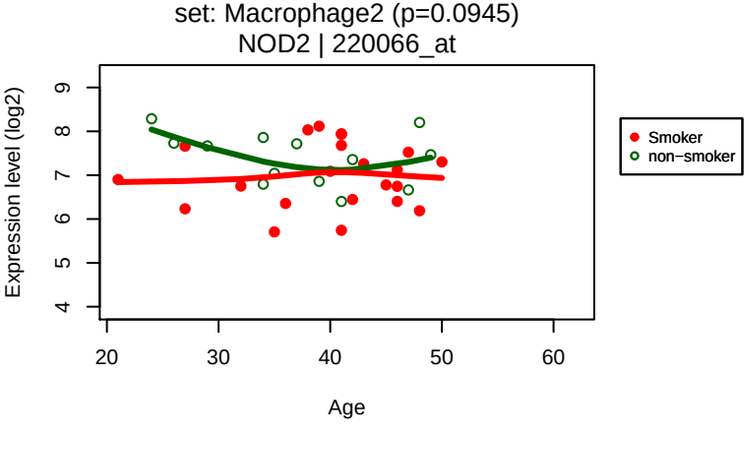

set: SmallAirway2 (p=0.0068)
NOD2 | 220066_at
Expression level (log2)
Age
Smoker
non−smoker

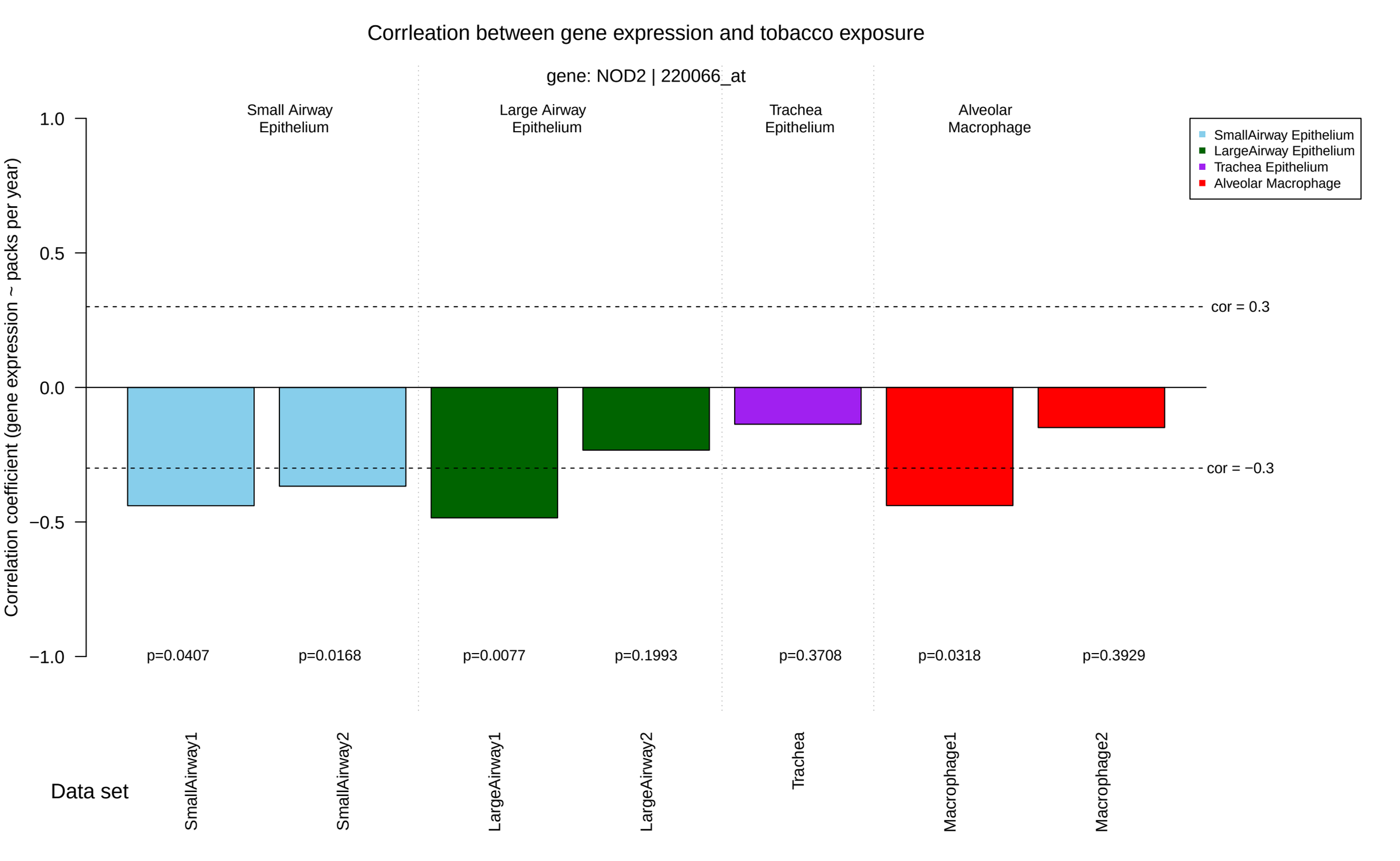

set: LargeAirway1 (p=0.0011)
NOD2 | 220066_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0087)
NOD2 | 220066_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0146)
NOD2 | 220066_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0066)
NOD2 | 220066_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0945)
NOD2 | 220066_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NOD2 | 220066_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0407
p=0.0168
p=0.0077
p=0.1993
p=0.3708
p=0.0318
p=0.3929
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2